STATIC PROGRAM ANALYSIS

Lecture 4 – Refactoring problems and AST limitations

ABSTRACT SYNTAX TREE

- Using visitor pattern can find more accurate answer to decision questions
- Knowing AST structure helps with answering question
  - Using a block visitor instead of visitors for each type of statement to count number of lines

SE REFACTORING PROBLEM

Renaming a given program variable

1. Problem statement
2. Rephrase it as a decision problem
3. Instantiate a SA
   - Method scope
   - Variable scope
   - Field of a class
4. Answer the decision problem

VARIABLE RENAMING

A 2-parts problem

1. Finding nodes to be renamed
2. Modifying the nodes

ECLIPSE AST MODIFIER/REWRITER

Rewriter is a special class ASTRewriter that takes in the ast to be modified.

```
CompilationUnit cu = (CompilationUnit) parser.createAST(null);
ast = cu.getAST();
rewriter = ASTRewrite.create(ast);
```

The same as for the AST visitor

Three way to make changes:

- Modifying a node: rewriter.replace(node, newNode, null);
- Removing a node: rewriter.remove(node, null)
- Adding a node:
  - ListRewrite listRewrite = rewriter.getListRewrite(node, CompilationUnit.*IMPORTS_PROPERTY*);
    listRewrite.insertFirst(newNode, null);

Changes depending on the type of node

Let's look at Rewriting.java

ANALYSIS DESIGN

Let's sketch it together.

For a given method m_i, a given block where the variable is defined b_i, change variable's name from s_1 to s_2. (Left as a HW 1 assignment)

SE CODE QUALITY PROBLEM

All variables must be initialized at the same time as they are declared.

- Instead of yellow warning, let's find those statements that are not initialized.
- Decision problem?
- Analysis design?

SE QUALITY PROBLEM

All assignments to variables must be used somewhere

- Somewhat challenging to do it over AST – need to remember way too much information
- Easier way is to propagate this information over a control flow graph CFG
- Reaching definition – a classic data-flow analysis
- WALA builds a CFG for you

NEXT TIME

WALA set-up

Obtaining CFG and iterating over it.

Reaching definitions analysis